Faculty Initial: TAJO
Dept. of CSE
School of Engineering
Date: Jan 29, 2026 12:00 PM

Examination	: Final Examination	**Semester**	: Fall 2025
Course Code	: CSE 263	**Marks**	: 30
Course Title	: Digital Logic Design	**Section**	: 5,6,7,9
Slot	: Slot - 14	**Duration**	: 2 hours

General Instructions:

- All questions must be answered.
- All the devices (mobile, smartwatch, etc) should be turned off and kept in the student's bag during the exam period.

Sl. No.	Tasks	Marks	COs	BCL*
1.	a) **Explain** why BCD addition cannot be performed using only a single 4-bit binary adder. Provide necessary truth tables and circuit diagrams to justify your answer.	4	CO2	L-2...
	b) **Construct** the function F(A,B,C,D) = $\sum m(2,3,4,5,8,9,12)$ using one 8:1 Mux.	3	CO2	L-3...
	c) **Demonstrate** how a 4-to-16 line decoder can be implemented using only 2-to-4 line decoders. 15863724-8	3	CO2	L-3...
2.	a) A system requires a JK flip-flop, but only a D flip-flop is available. **Derive** the D input equation and sketch the logic diagram to realize JK flip-flop behavior using the D flip-flop.	4 246135	CO3	L-4...
	b) A designer uses a JK flip-flop in a counter but notices multiple output transitions within a single clock pulse. **Explain** why this occurs and demonstrate a circuit-level solution to ensure only one state change per clock cycle. 1357246	4	CO3	13529 L-4...
3.	a) A controller cycles through 11 unique states and then automatically resets to the initial state. The system is built using asynchronously connected JK flip-flops, where unused output combinations trigger a reset. 158637924 **Identify** the required counter type and justify the use of JK flip-flops instead of SR flip-flops.	4	CO3	L-4...

*Bloom's Cognitive Level

	b) A control unit cycles through eight states, advancing one state per clock pulse, with the requirement that all state transitions occur simultaneously on the same clock edge to avoid timing errors. **Analyze** the requirement to identify the appropriate counter, and illustrate its logic diagram and state transition diagram.	4	CO3	L-4...
4.	A digital system captures a 5-bit input simultaneously and, under control of a mode signal, later transmits the stored data serially over a single output line using a clock; **analyze** the operation of the system in both modes, draw the required circuit using flip-flops, and for the input 10101 sketch the timing diagram showing the serial output assuming MSB is transmitted first.	4	CO4	L-4...

0 1 0 1 0 1 0 1
0 0 1 1 0 0 1 1
0 0 0 0 1 1 1 1

*Bloom's Cognitive Level